Tobias Mwangi-Hartley

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PROFESSIONAL SUMMARY

Data Scientist with experience building and evaluating forecasting, churn and experimentation solutions using Python, SQL and cloud data tools. Delivered scheduled forecasting workflows and weekly batch churn scoring, with measurable business outcomes. Brings a strong statistical foundation and practical experience in feature engineering, model evaluation and automated testing, and is developing further expertise in production machine-learning engineering.

TECHNICAL SKILLS

Programming and data: Python, SQL, R

Machine learning: LightGBM, XGBoost, scikit-learn, SHAP, feature engineering, forecasting, churn modelling, model evaluation

Data and cloud: BigQuery, Vertex AI Workbench, dbt

Engineering and applications: pytest, Docker, Streamlit, R Markdown

Additional: Looker, statsmodels, PyTorch (course project)

PROFESSIONAL EXPERIENCE

Data Scientist | Thistlebox *2022–Present*

- Developed a weekly recipe-demand forecasting workflow producing six-week-ahead forecasts by recipe and region, using LightGBM and calendar, price and menu-position features.
- Implemented the forecasting workflow as a scheduled notebook in Vertex AI Workbench, writing outputs to BigQuery for operational use in Looker.
- Improved forecast WAPE from 23% to 14%; the work was associated with an estimated 9% reduction in food waste.
- Built a weekly XGBoost churn-scoring workflow in BigQuery to identify top-decile customers for retention offers. A 10% holdout evaluation found churn was 2.1 percentage points lower in the treated group.
- Built a Streamlit application for planning and analysing sequential A/B tests, used by approximately 25 people in product and growth. Wrote its Dockerfile; the data platform team handled deployment to Cloud Run.
- Began writing pytest unit tests for forecasting feature code in 2025; test coverage is approximately 60%.
- Delivered the internal “Stats for PMs” training three times.

Junior Data Analyst | Firth Energy Retail *2020–2022*

- Segmented customers and analysed tariffs using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team approximately two days per month.

EDUCATION

MSc in Statistics

- Dissertation on Bayesian hierarchical models.
- Academic foundation in experimental design and modelling uncertainty.

SELECTED PROJECT

Image Classification Course Project | Personal *2025*

- Fine-tuned a small CNN on a public food-images dataset as part of an online deep-learning course, achieving approximately 80% accuracy.
- Tools: PyTorch, torchvision, Google Colab.